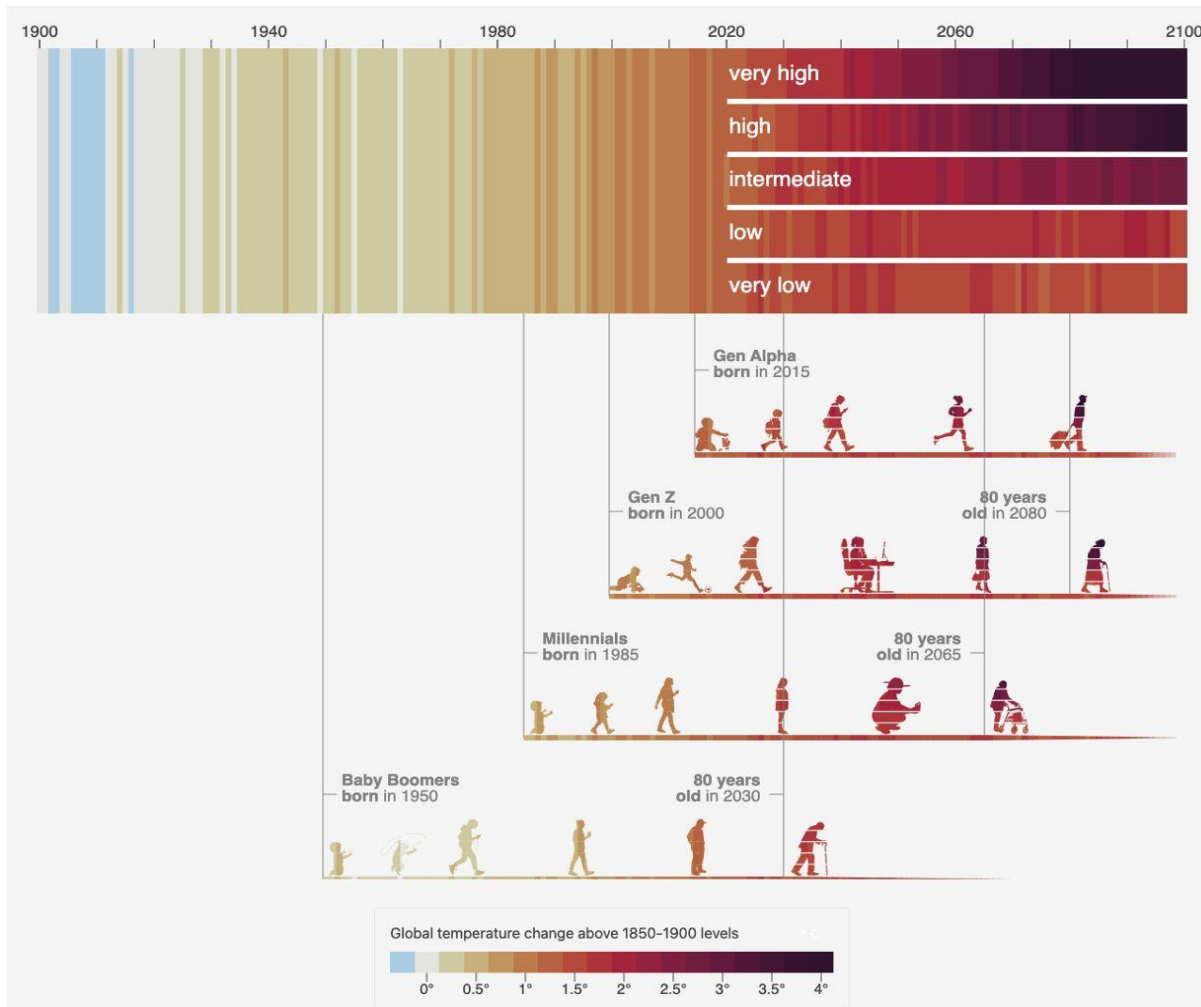


Climate Change Governance: Science, Data & Models



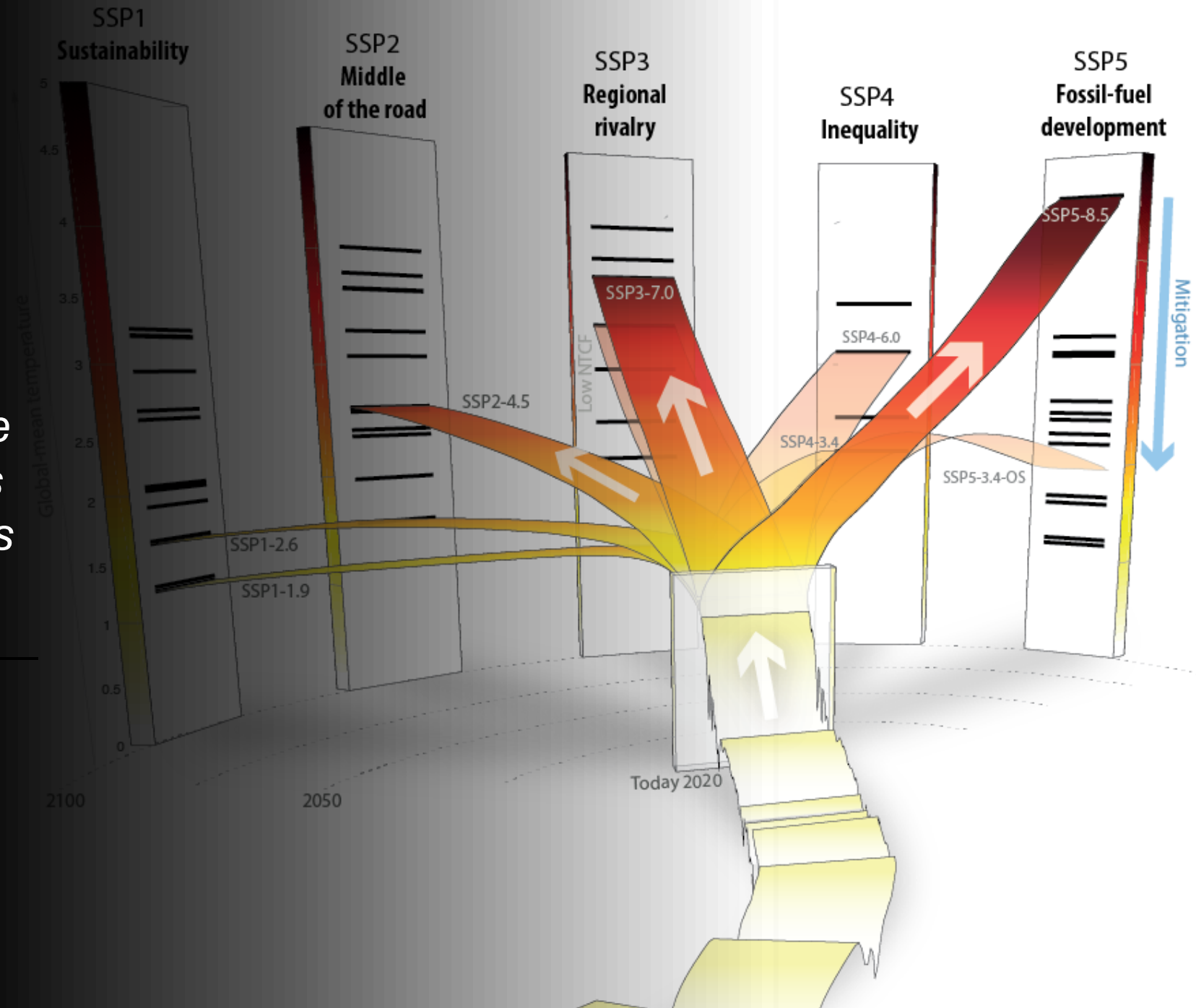
Lecture 8: SSPs

ENGG562/ENV562 – Fall 2026

Muhammad Awais

Shared Socioeconomic Pathways (SSPs)

*RCPs told us **how much** the climate might change; SSPs tell us **what kind of world** is experiencing that change.*



SSPs aim to fill four haps

The same forcing (RCP) can arise in very different societies. The SSPs attempt to describe those societies.



Socioeconomic context

Each RCP came from a different IAM, so the worlds behind them were never comparable.



Impacts

Vulnerability depends on sensitivity and adaptive capacity, and both are social aspects rather than physical.



Mitigation feasibility

Whether 2.6 W/m^2 is reachable depends on the society that is trying to reach it.



Policy

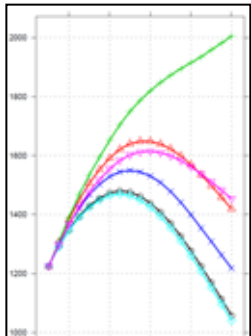
RCPs could not separate a world without climate policy from what policy then achieves.

Shared Socioeconomic Pathways (SSPs)



Narrative

Qualitative description
of broad patterns of
development
Logic relating elements
of narrative to each
other



Quantitative elements

National:
Population
Education
Urbanization
GDP

SSPs

- 1: Sustainability
- 2: Middle of the Road
- 3: Regional Rivalry
- 4: Inequality
- 5: Fossil-fueled Development

SSP narratives, quantitative elements: **2017 special issue of *Global Environmental Change*.**

SSP Database, hosted by IIASA.

Basic Elements and IAM Scenarios for the SSP (GEC, 2017)

- Community-wide effort
 - Economists
 - Demographers
 - Impact & Vulnerability
 - Integrated Assessment Modellers
- **Global Environmental Change**
 - Overview (Riahi et al. 2017)
 - Demographic projections (KC & Lutz 2017)
 - GDP projections (OECD, IIASA, PIK 2017)
 - Urbanisation projections (Liang & O'Neill 2017)
 - Quantifications of SSPs (6 global IAM teams)
 - Cross-cutting papers on energy, land and air pollution



Contents lists available at ScienceDirect

Global Environmental Change

journal homepage: www.elsevier.com/locate/gloenvcha



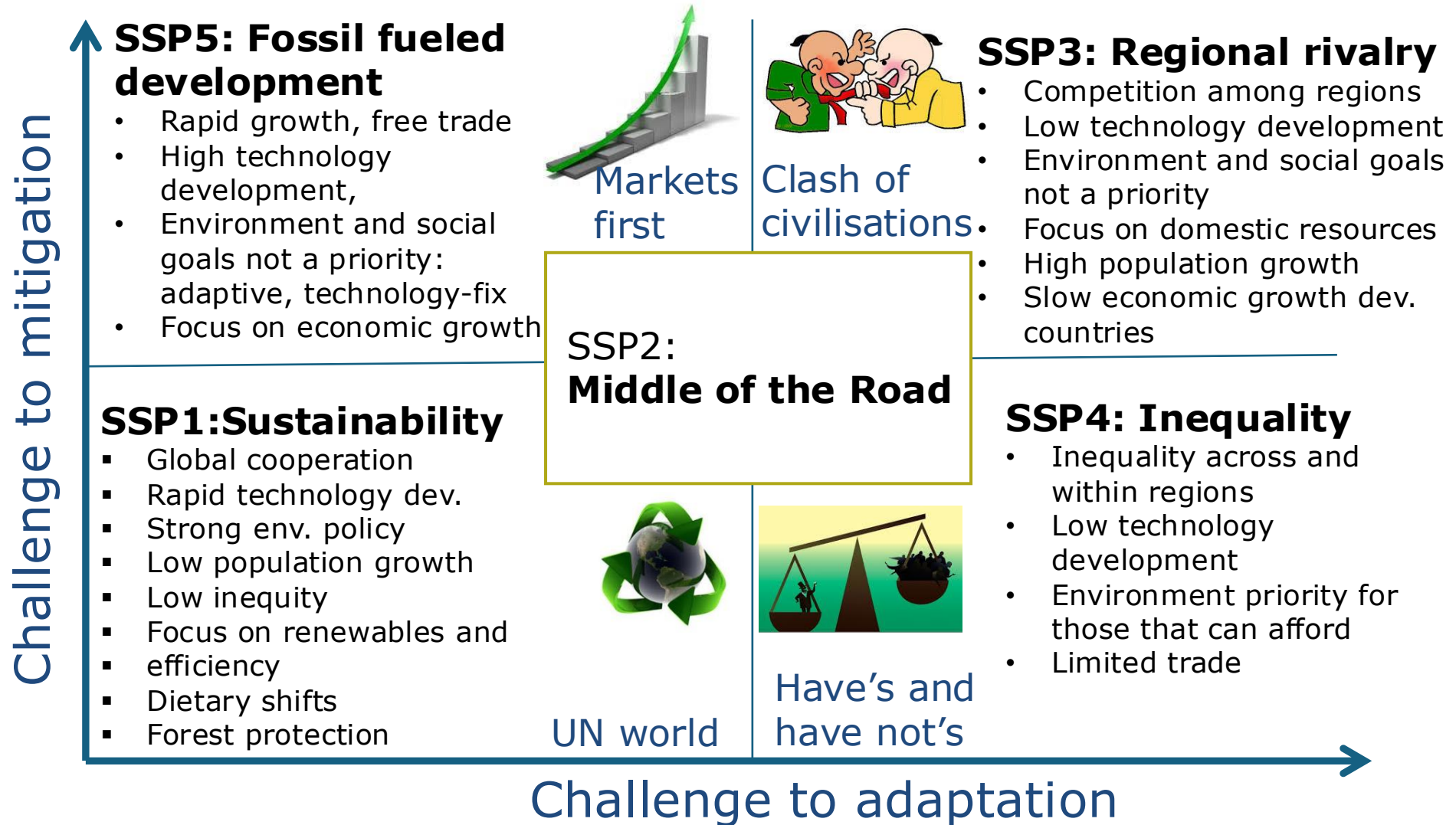
The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview

Keywan Riahi^{a,*}, Detlef P. van Vuuren^b, Elmar Kriegler^c, Jae Edmonds^d, Brian C. O'Neill^e, Shinichiro Fujimori^f, Nico Bauer^c, Katherine Calvin^d, Rob Dellink^g, Oliver Fricko^a, Wolfgang Lutz^a, Alexander Popp^c, Jesus Crespo Cuaresma^a, Samir KC^{a,h}, Marian Leimbach^c, Leiwen Jiang^e, Tom Kram^b, Shilpa Rao^a, Johannes Emmerling^{i,j}, Kristie Ebi^k, Tomoko Hasegawa^l, Petr Havlik^a, Florian Humpenöder^c, Lara Aleluia Da Silva^{i,j}, Steve Smith^d, Elke Stehfest^b, Valentina Bosetti^{i,j,l}, Jiyong Eom^{d,m}, David Gernaat^b, Toshihiko Masui^f, Joeri Rogelj^a, Jessica Strefler^c, Laurent Drouet^{i,j}, Volker Krey^a, Gunnar Luderer^c, Mathijs Harmsen^b, Kiyoshi Takahashi^f, Lavinia Baumstark^c, Jonathan C. Doelman^b, Mikiko Kainuma^f, Zbigniew Klimont^a, Giacomo Marangoni^{i,j}, Hermann Lotze-Campen^{c,p}, Michael Obersteiner^a, Andrzej Tabeauⁿ, Massimo Tavoni^{i,j,o}

Data available
at the SSP database
hosted by IIASA

<https://data.ece.iiasa.ac.at/ssp/#/workspaces>

The Scenario Matrix Architecture

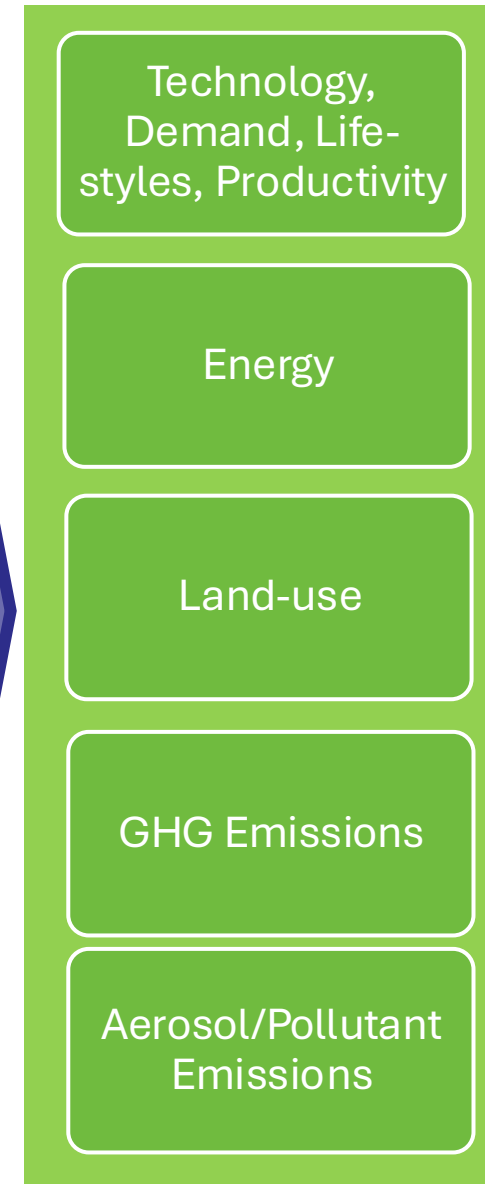
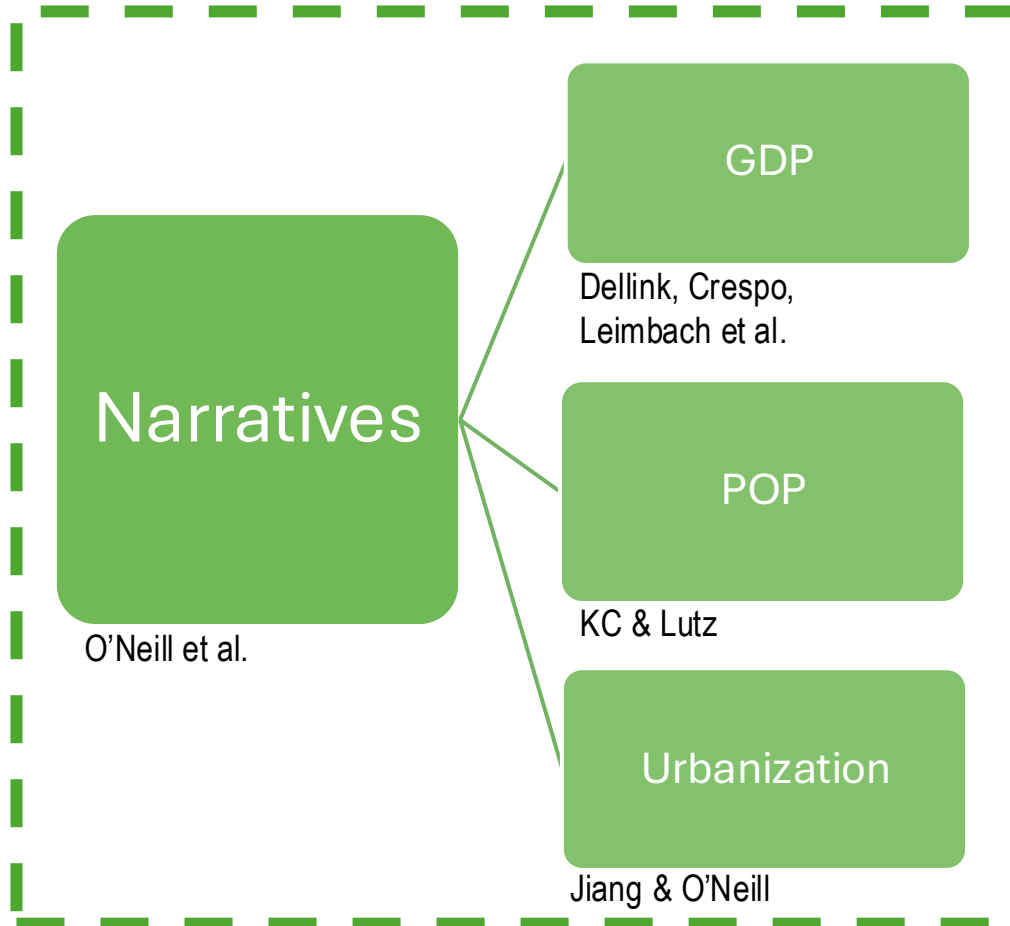


Which SSP trajectory do you think the world is currently on ?

SSP Quantifications

SSP interpretations by IAMs

SSPs (Basic Elements/Drivers)



AIM/CGE, GCAM, IMAGE, MESSAGE-GLOBIOM, REMIND-MAGPIE, WITCH-GLOBIOM

SSP-based emissions projections

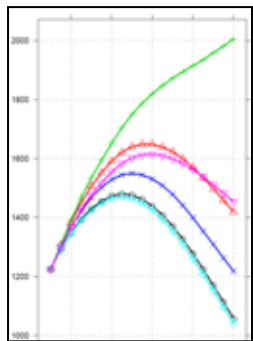
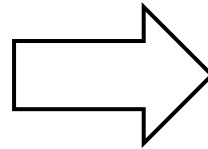
SSPs 1-5

Emissions



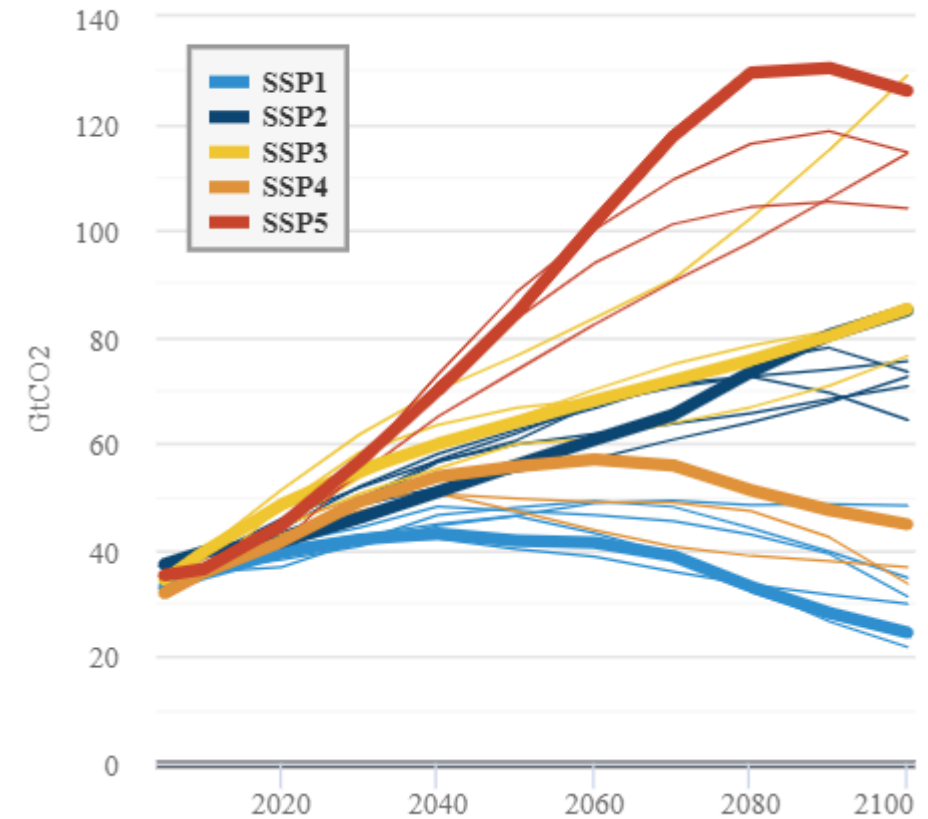
Narratives

Integrated
Assessment
Models

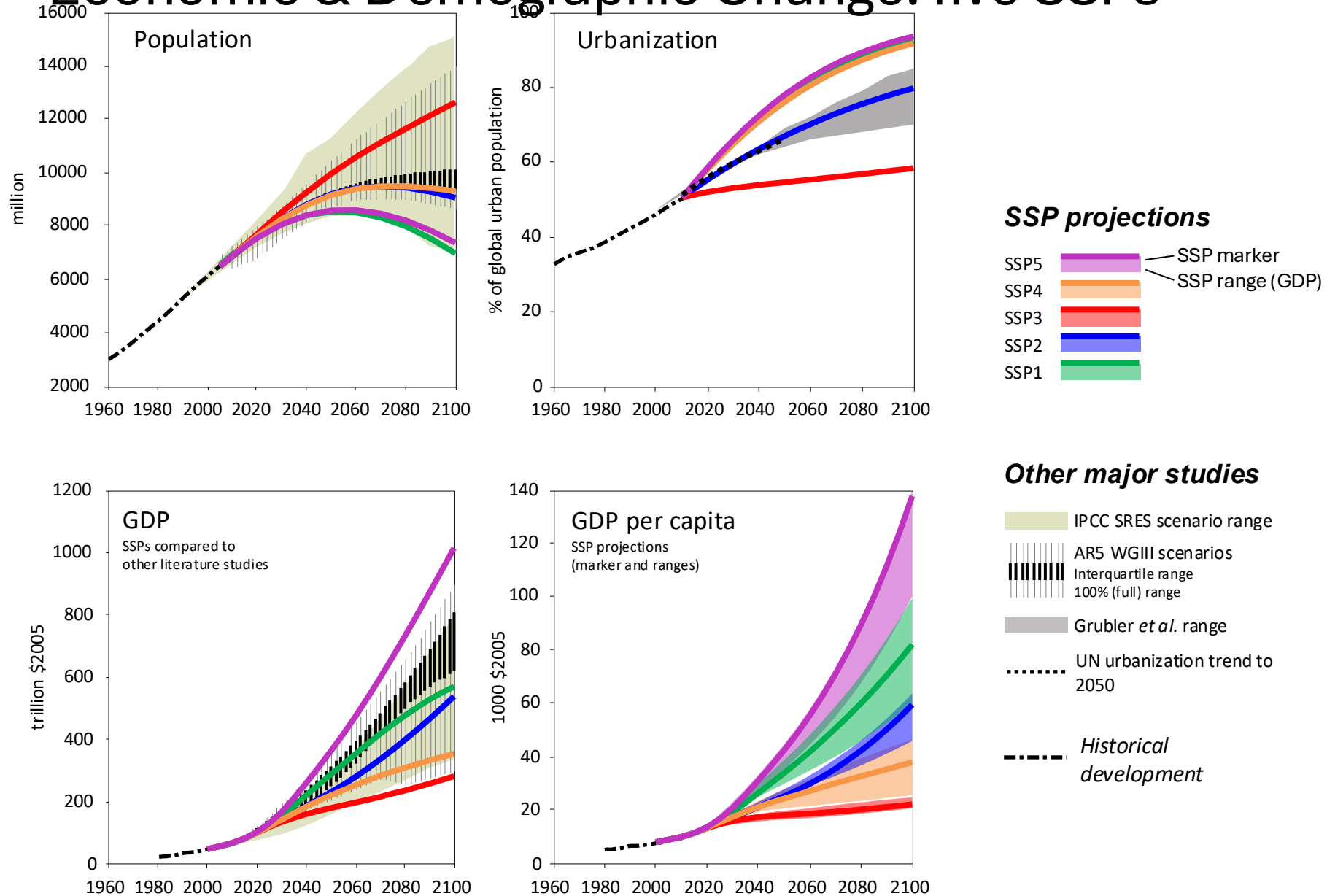


Quantitative
Elements

CO2 emissions for SSP baselines



Economic & Demographic Change: five SSPs



Sources: Overview: Riahi et al, 2017, GDP: Dellink et al, 2017; Urbanization: Jiang and O'Neill et al, 2017, Population: KC and Lutz, 2017

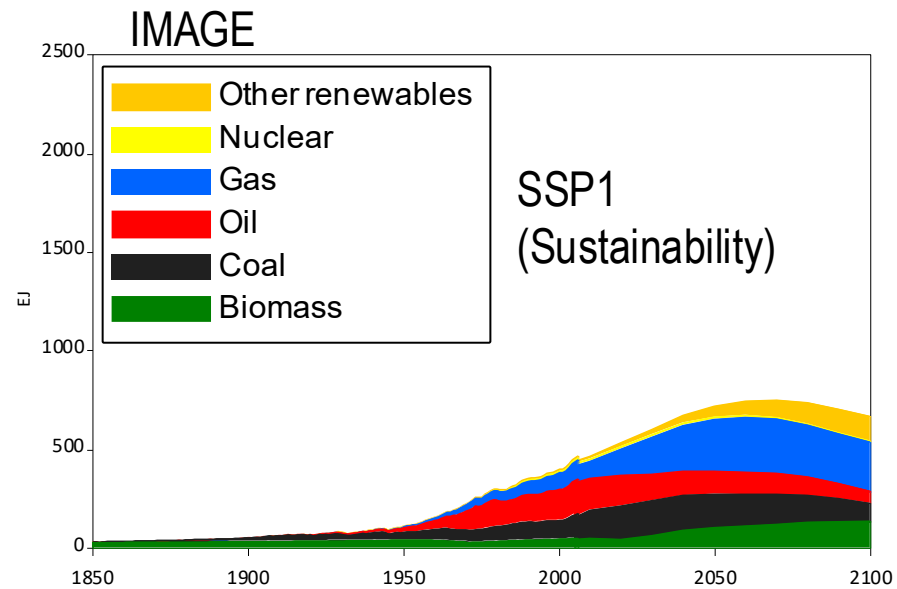
Reference SSP (IAM) Scenarios

(no climate policy beyond those in place today)

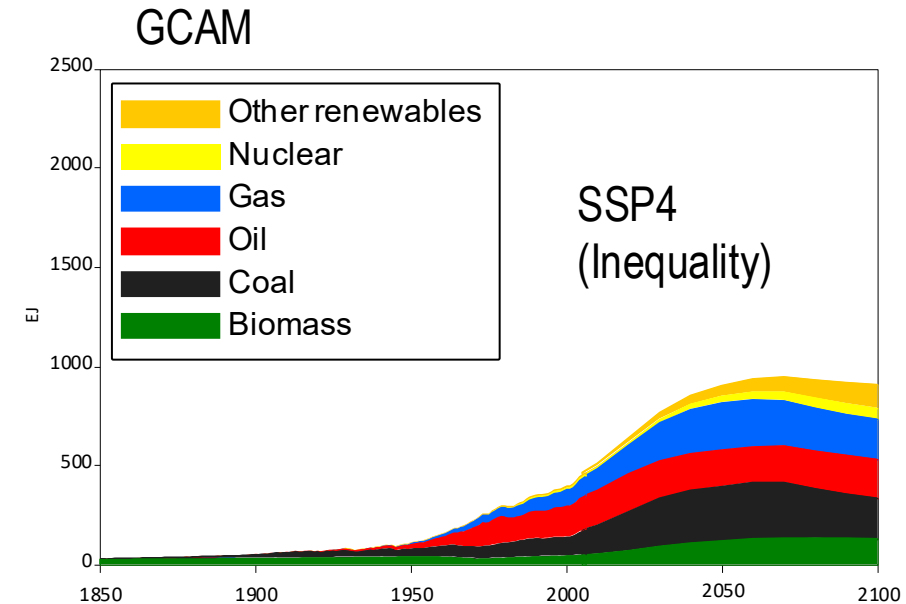
- Six IAM teams
- Five SSPs
- One representative Marker Scenario for each SSP
- For each SSP there are multiple IAM runs depicting uncertainty ranges

Energy – SSP Reference Cases

Two marker scenarios where mitigation is relatively easy



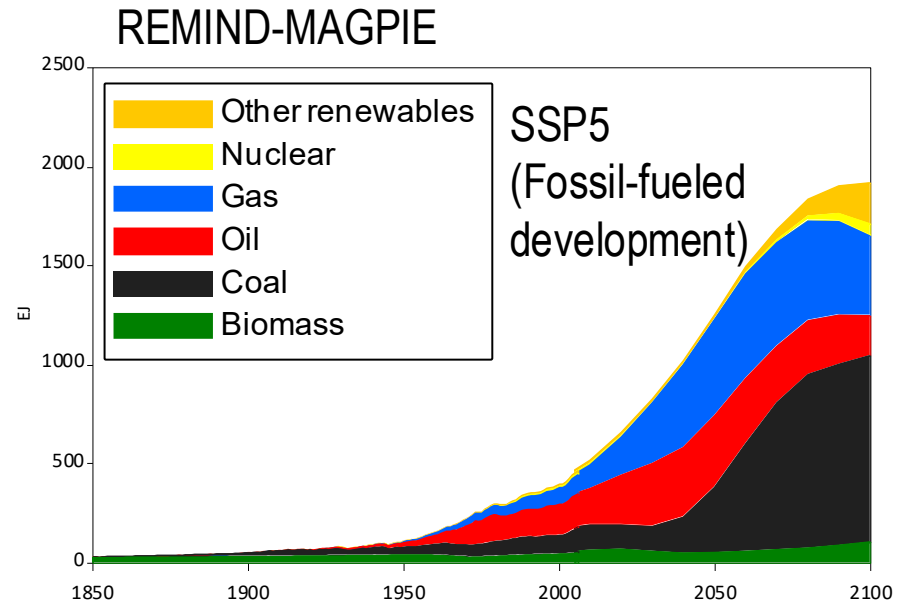
Transition away from coal/oil
Low demand



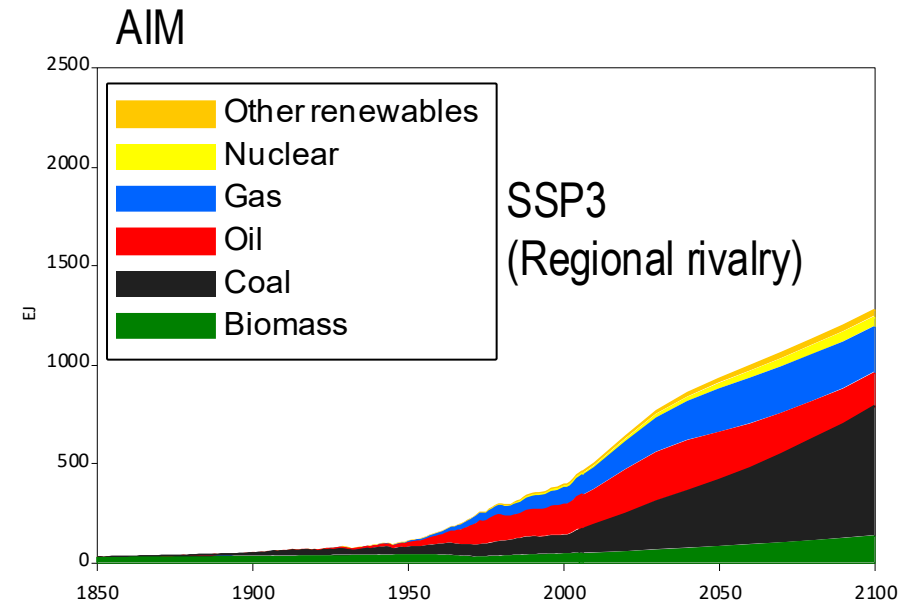
High share of poor with low emissions
Low/intermediate demand
Technology available to the “elite”

Energy – SSP Reference Cases

Two marker scenarios where mitigation is relatively difficult



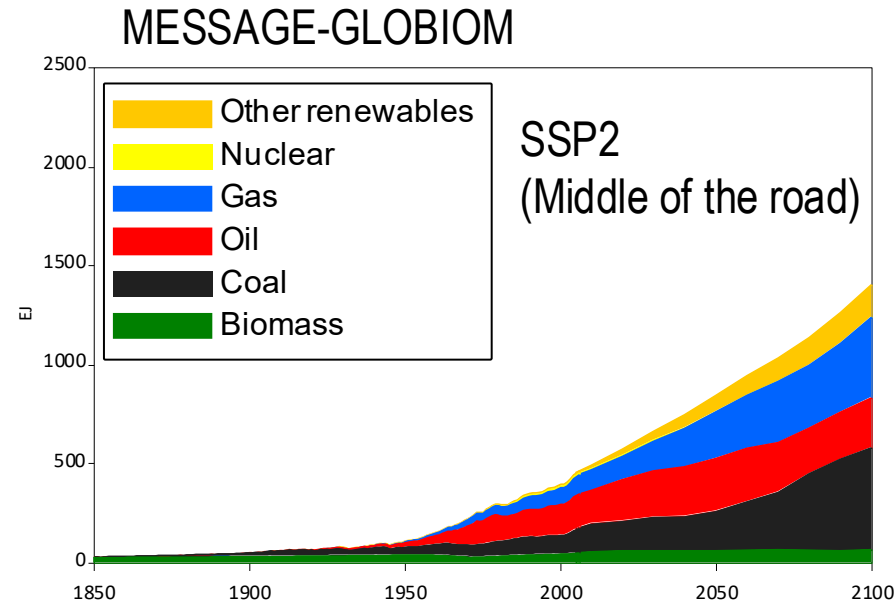
Coal-intensive development
Very high demand



Fossil-intensive
High poverty
Slow technological change
Strong fragmentation

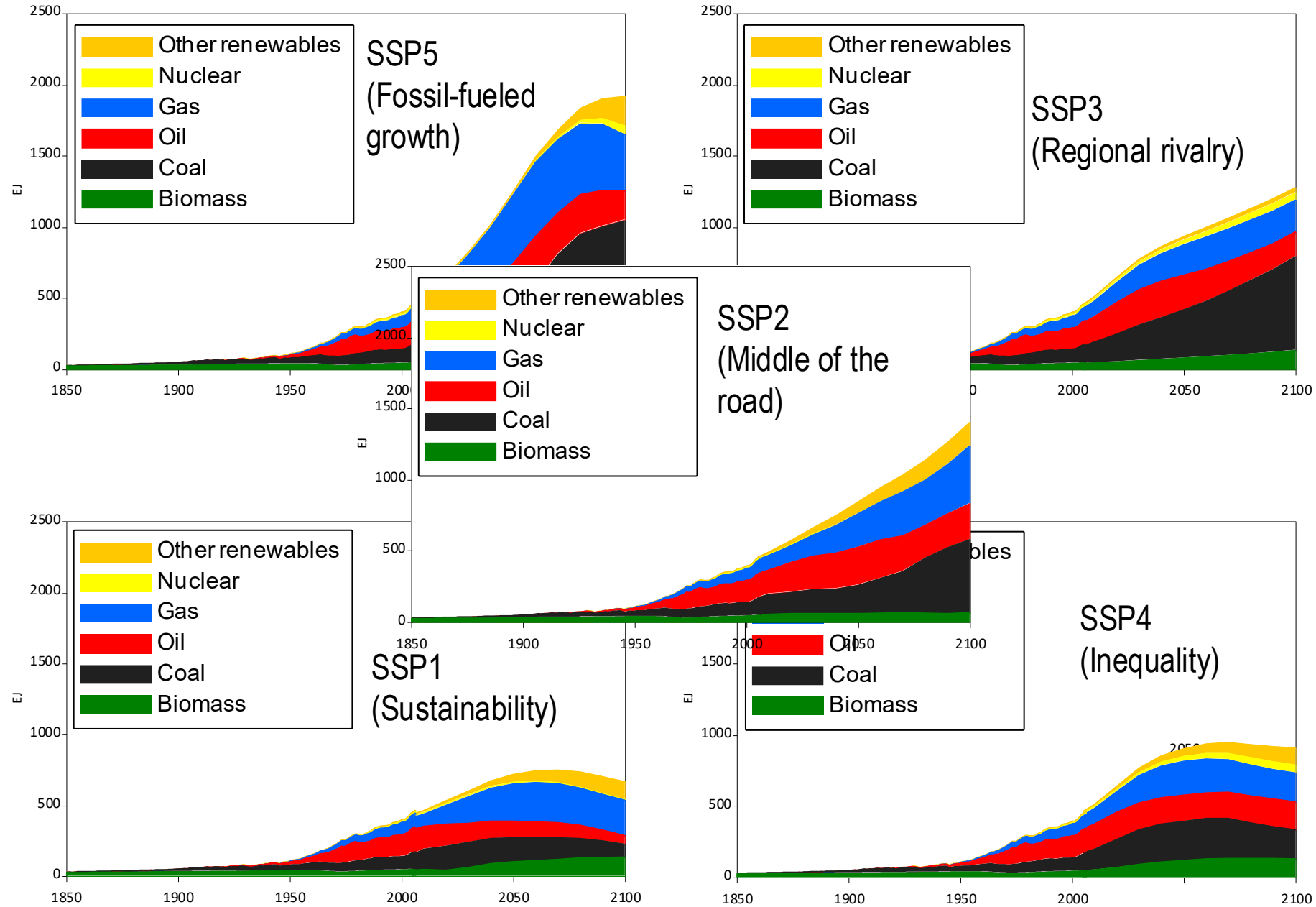
Energy – SSP Reference Cases

A central marker scenario with intermediate mitigation challenge



Balanced Technology
Intermediate demand

Primary Energy – SSP Reference Cases



How were these pathways created?

Storylines

SSP5: Fossil fueled development

- Rapid growth, free trade
- High technology development,
- Environment and social goals not a priority: adaptive, technology-fix
- Focus on economic growth

SSP1:Sustainability

- Global cooperation
- Rapid technology dev.
- Strong env. policy
- Low population growth
- Low inequity
- Focus on renewables and efficiency
- Dietary shifts
- Forest protection

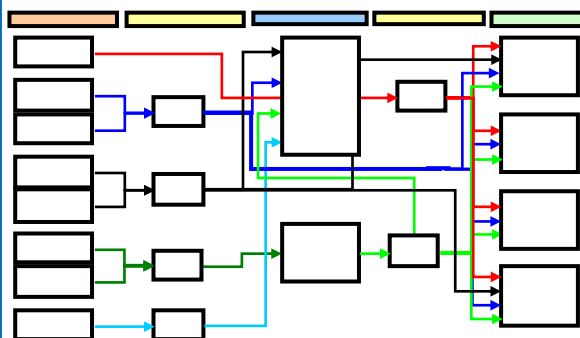
Qualitative assumptions

Table A.1: Qualitative assumptions for energy demand across SSPs

SSP Element	SSP 1			SSP 2			SSP 3			SSP 4			SSP 5		
	Country Income Groupings														
	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High
Non-climate Policies															
Traditional Fuel Use	fast phase-out, driven by policies and economic development			intermediate phase-out, regionally diverse speed			continued reliance on traditional fuels			continued traditional fuel use among low income households			fast phase-out, driven by development priority		
Energy Demand Side															
Lifestyles	modest service demands (less material intensive)			medium service demands (generally material intensive)			medium service demands (material intensive)			low service demands			modest service demands		
Environmental Awareness	high			medium			low			low			high		
Energy Intensity of Services															
Industry	low			medium			high			high			low		
Buildings	low			medium			high			medium			low/medium		
Transportation	low			medium			medium			high			low		
General Comments				some regional diversity retained											

Community

IAM

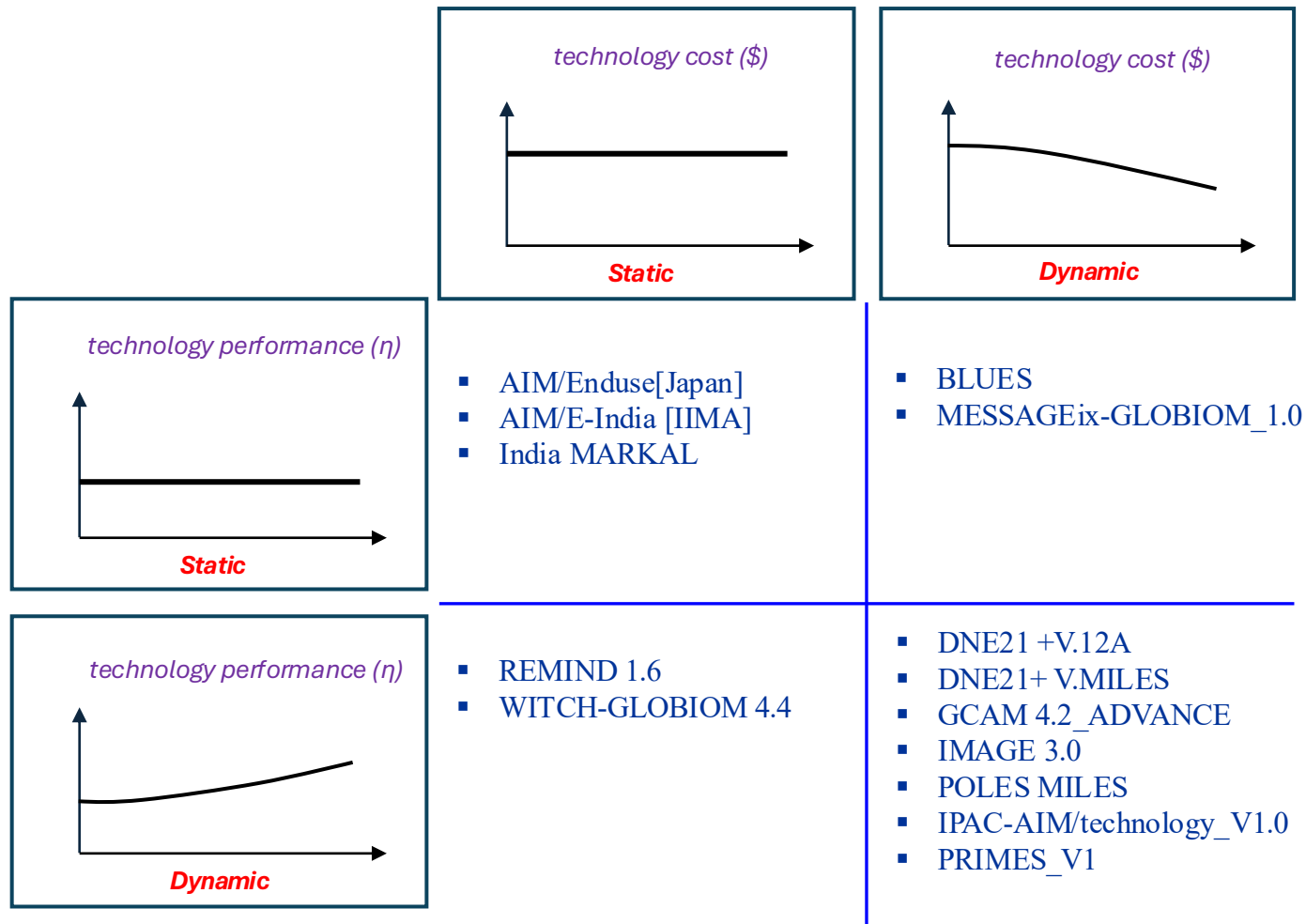


Modeling Team

Quantitative assumptions

MODEL	SCENARIO	REGION	VARIABLE	UNIT	2010	2015	2020	2025	2030	2035	2040	2045	2050
2	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Biomass	US\$2010/kWe	2284.664	2284.664	2284.664	2284.664	2284.664	2284.664	2284.664	2284.664
3	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Coal/IGCC	US\$2010/kWe	2263.432	2263.432	2263.432	2263.432	2263.432	2263.432	2263.432	2263.432
4	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] CSP	US\$2010/kWe	7189.384	7189.384	7189.384	7189.384	7189.384	7189.384	7189.384	7189.384
5	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Gas/CC	US\$2010/kWe	1276.799	1276.799	1276.799	1276.799	1276.799	1276.799	1276.799	1276.799
6	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Gas/CT	US\$2010/kWe	526.7987	526.7987	526.7987	526.7987	526.7987	526.7987	526.7987	526.7987
7	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Hydro	US\$2010/kWe	3074.07	3074.07	3074.07	3074.07	3074.07	3074.07	3074.07	3074.07
8	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Nuclear	US\$2010/kWe	4086.623	4086.623	4086.623	4086.623	4086.623	4086.623	4086.623	4086.623
9	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] PV	US\$2010/kWe	4648.405	4648.405	4648.405	4648.405	4648.405	4648.405	4648.405	4648.405
10	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Wind/Onshore	US\$2010/kWe	3103.509	3103.509	3103.509	3103.509	3103.509	3103.509	3103.509	3103.509
11	AIM/E-India (IIMA)	Reference	India	Capital Cost [Electricity] Wind/Offshore	US\$2010/kWe	1693.652	1693.652	1693.652	1693.652	1693.652	1693.652	1693.652	1693.652
12	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Biomass	US\$2010/kWe/yr	22.82728	22.82728	22.82728	22.82728	22.82728	22.82728	22.82728	22.82728
13	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Coal/IGCC	US\$2010/kWe/yr	44.64246	44.64246	44.64246	44.64246	44.64246	44.64246	44.64246	44.64246
14	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Coal/PC	US\$2010/kWe/yr	44.64246	44.64246	44.64246	44.64246	44.64246	44.64246	44.64246	44.64246
15	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] CSP	US\$2010/kWe/yr	117.8658	117.8658	117.8658	117.8658	117.8658	117.8658	117.8658	117.8658
16	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Gas/CC	US\$2010/kWe/yr	13.69637	13.69637	13.69637	13.69637	13.69637	13.69637	13.69637	13.69637
17	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Gas/CT	US\$2010/kWe/yr	11.07591	11.07591	11.07591	11.07591	11.07591	11.07591	11.07591	11.07591
18	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Hydro	US\$2010/kWe/yr	35.48955	35.48955	35.48955	35.48955	35.48955	35.48955	35.48955	35.48955
19	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Nuclear	US\$2010/kWe/yr	99.52695	99.52695	99.52695	99.52695	99.52695	99.52695	99.52695	99.52695
20	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] PV	US\$2010/kWe/yr	105.0605	105.0605	105.0605	105.0605	105.0605	105.0605	105.0605	105.0605
21	AIM/E-India (IIMA)	Reference	India	OM Cost [Fixed] [Electricity] Wind/Onshore	US\$2010/kWe/yr	50.47305	50.47305	50.47305	50.47305	50.47305	50.47305	50.47305	50.47305
22	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	OM Cost [Fixed] [Electricity] Wind/Onshore	US\$2010/kWe/yr	59.50295	59.50295	59.50295	59.50295	59.50295	59.50295	59.50295	59.50295
23	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	Capital Cost [Electricity] Biomass	US\$2010/kWe	3524.209	4007.529	4007.529	4007.529	4007.529	4007.529	4007.529	4007.529
24	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	Capital Cost [Electricity] Coal/IGCC	US\$2010/kWe	2894.886	2894.886	2894.886	2894.886	2894.886	2894.886	2894.886	2894.886
25	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	Capital Cost [Electricity] Coal/PC11	US\$2010/kWe	2498.782	2498.782	2498.782	2498.782	2498.782	2498.782	2498.782	2498.782
26	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	Capital Cost [Electricity] Coal/PC12	US\$2010/kWe	2517.292	2517.292	2517.292	2517.292	2517.292	2517.292	2517.292	2517.292
27	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	Capital Cost [Electricity] Gas/CC	US\$2010/kWe	1213.256	1213.256	1213.256	1213.256	1213.256	1213.256	1213.256	1213.256
28	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	Capital Cost [Electricity] Gas/CT	US\$2010/kWe	1547.214	1547.214	1547.214	1547.214	1547.214	1547.214	1547.214	1547.214
29	AIM/Enduse[Japan]	JPN_MILE2_INDC80	Japan	Capital Cost [Electricity] Gas/CC	US\$2010/kWe	1152.593	1152.593	1152.593	1152.593	1152.593	1152.593	1152.593	1152.593

Projecting techno-economic parameters



(using coal power plants as the example)

Qualitative Assumptions: Demand

Table A.1: Qualitative assumptions for energy demand across SSPs

SSP Element	SSP 1			SSP 2			SSP 3			SSP 4			SSP 5								
	Country Income Groupings																				
	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High						
Non-climate Policies																					
Traditional Fuel Use	fast phase-out, driven by policies and economic development			intermediate phase-out, regionally diverse speed			continued realiance on traditional fuels			continued traditional fuel use			some traditional fuel use among low income houtholds			fast phase-out, driven by development priority					
Energy Demand Side																					
Lifestyles	modest service demands (less material intensive)			medium service demands (generally material intensive)			medium service demands (material intensive)			low service demands			modest service demands			high service demands (very material intensive)					
Environmental Awareness	high			medium			low			low			high			medium (low for global level/high for local level)					
Energy Intensity of Services																					
Industry	low			medium			high			high			low			medium					
Buildings	low			medium			high			medium			low/medium			medium					
Transportation	low			medium			medium			high			low/medium			low			high		
General Comments				some regional diversity retained																	

Qualitative Assumptions: Fossil Fuels

Table A.2: Qualitative assumptions for fossil energy supply across SSPs

	SSP1 Sustainability	SSP2 Middle of the Road	SSP3 Regional Rivalry Country grouping Exporter Importer	SSP4 Inequality Country grouping by income Low Medium High	SSP5 Fossil fueled development
Coal					
Macro-economy	cost driver	neutral	cost reducing	cost driver cost driver neutral	cost reducing
Technological progress	slow	medium	slow fast	medium	very fast
National & environmental policy	very restrictive	supportive	very supportive	supportive supportive restrictive	very supportive
Conv. hydrocarbons					
Macro-economy	neutral	neutral	neutral	cost driver neutral cost reducing	cost reducing
Technological progress	medium	medium	medium	fast	very fast
National & environmental policy	restrictive	supportive	not supportive supportive	supportive supportive restrictive	very supportive
Unconv. hydrocarbons					
Macro-economy	neutral	neutral	neutral	cost driver neutral cost reducing	cost reducing
Technological progress	slow	medium	slow medium	medium	very fast
National & environmental policy	very restrictive	supportive	not supportive very supportive	supportive supportive restrictive	very supportive
General					
Trade barriers	free trade	some barriers	high barriers	barriers	free

Qualitative Assumptions: Conversion

Table A.3: Qualitative assumptions for energy conversion technologies SSPs

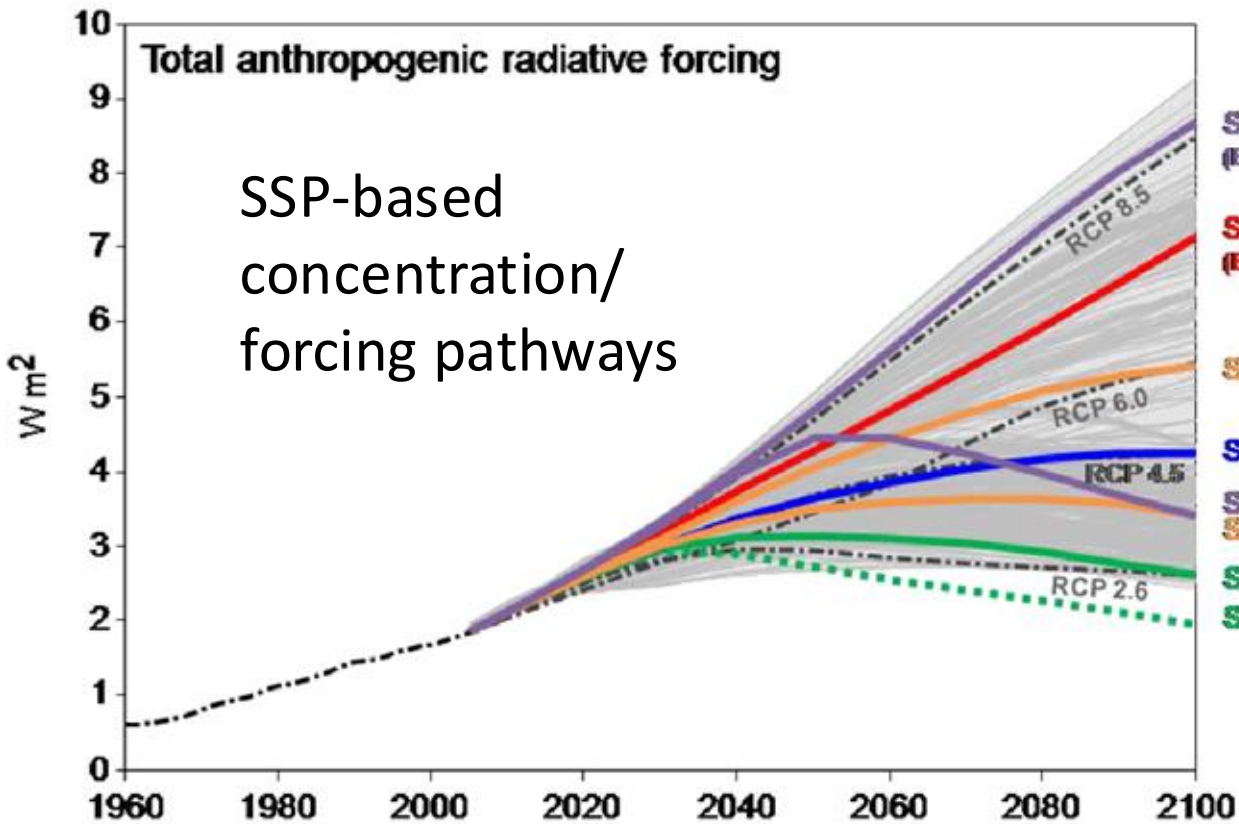
SSP Element	SSP 1			SSP 2			SSP 3			SSP 4			SSP 5		
	Country Income Groupings														
	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High
Conventional and Unconventional Fossil Fuel Conversion (synfuel and syngas in parenthesis if different)															
Technology Development	Med			Med			Low			Low	Med	Med	Med (High)		
Social Acceptance	Low			Med			High			High	Low	Low	High		
Commercial Biomass Conversion															
Technology Development	High			Med			Low			High	High	High	Med		
Social Acceptance	Low			Med			High			High	High	High	Med		
Non-bio Renewables Conversion															
Technology Development	High			Med			Low			High	High	High	Med		
Social Acceptance	High			Med			Med			High	High	High	Low		
Nuclear Power															
Technology Development	Med			Med			Low	Low	Med	High	High	High	Med		
Social Acceptance	Low			Med			High	High	High	High	Med	Med	Med		
CCS (under climate policy only)															
Technology Development	Med			Med			Med			High	High	High	High		
Social Acceptance	Low			Med			Med			High	Med	Med	Med		

SSP-based climate projections

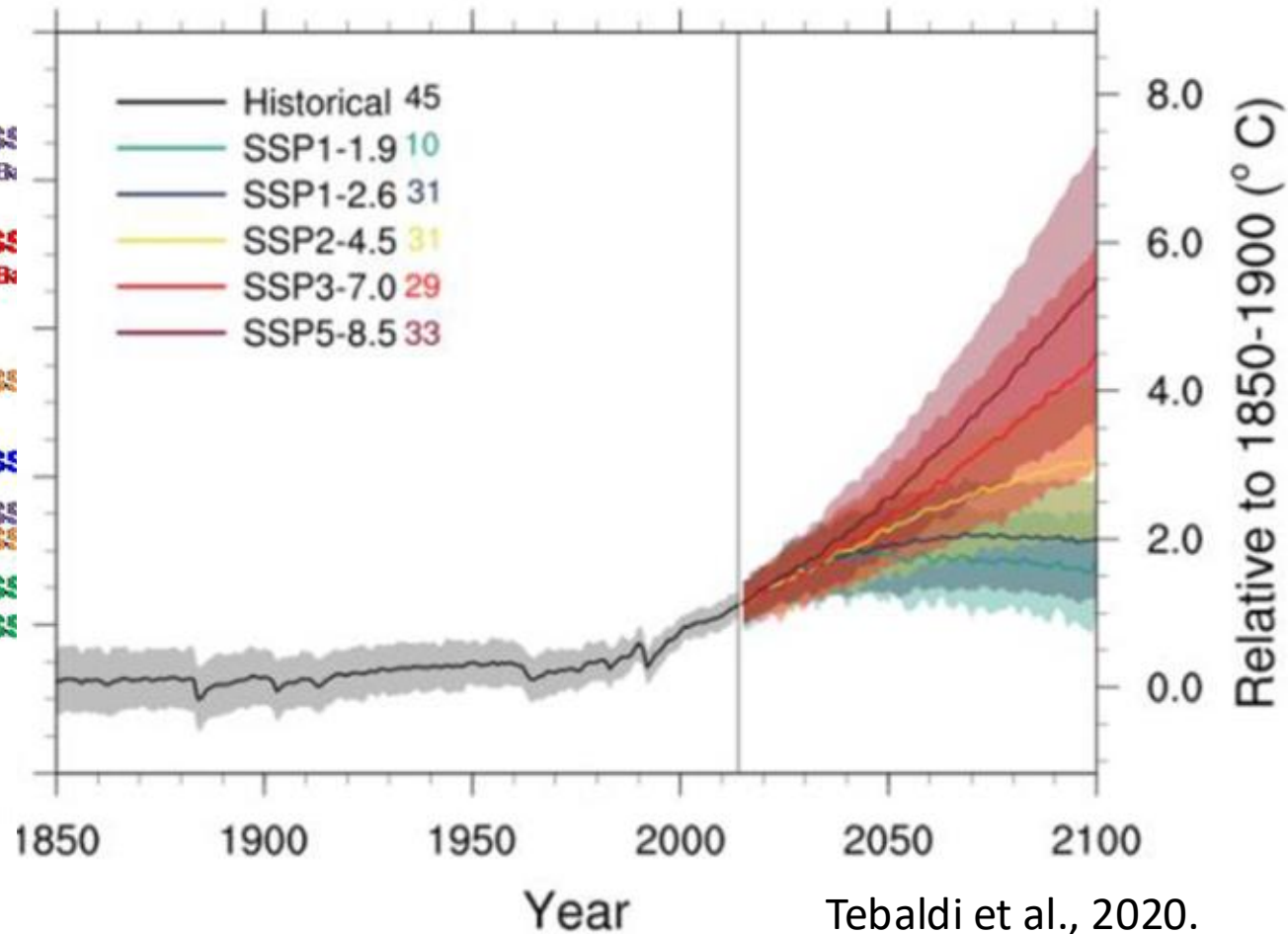
Radiative forcing
(from concentrations)



Global Mean Temp. Change
(CMIP6)



O'Neill et al., 2016; Riahi et al., 2017.



Tebaldi et al., 2020.

Qualitative Assumptions: Demand

Table A.1: Qualitative assumptions for energy demand across SSPs

SSP Element	SSP 1			SSP 2			SSP 3			SSP 4			SSP 5								
	Country Income Groupings																				
	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High						
Non-climate Policies																					
Traditional Fuel Use	fast phase-out, driven by policies and economic development			intermediate phase-out, regionally diverse speed			continued realiance on traditional fuels			continued traditional fuel use some traditional fuel use among low income housholds			fast phase-out, driven by development priority								
Energy Demand Side																					
Lifestyles	modest service demands (less material intensive)			medium service demands (generally material intensive)			medium service demands (material intensive)			low service demands			modest service demands			high service demands (very material intensive)					
Environmental Awareness	high			medium			low			low			high			medium (low for global level/high for local level)					
Energy Intensity of Services																					
Industry	low			medium			high			high			low			medium					
Buildings	low			medium			high			medium			low/medium			medium					
Transportation	low			medium			medium			high			low/medium			low			high		
General Comments				some regional diversity retained																	

Qualitative Assumptions: Fossil Fuels

Table A.2: Qualitative assumptions for fossil energy supply across SSPs

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Coal					
Macro-economy	cost driver	neutral	cost reducing	cost driver cost driver neutral	cost reducing
Technological progress	slow	medium	slow fast	medium	very fast
National & environmental policy	very restrictive	supportive	very supportive	supportive supportive restrictive	very supportive
Conv. hydrocarbons					
Macro-economy	neutral	neutral	neutral	cost driver neutral cost reducing	cost reducing
Technological progress	medium	medium	medium	fast	very fast
National & environmental policy	restrictive	supportive	not supportive supportive	supportive supportive restrictive	very supportive
Unconv. hydrocarbons					
Macro-economy	neutral	neutral	neutral	cost driver neutral cost reducing	cost reducing
Technological progress	slow	medium	slow medium	medium	very fast
National & environmental policy	very restrictive	supportive	not supportive very supportive	supportive supportive restrictive	very supportive
General					
Trade barriers	free trade	some barriers	high barriers	barriers	free

Qualitative Assumptions: Conversion

Table A.3: Qualitative assumptions for energy conversion technologies SSPs

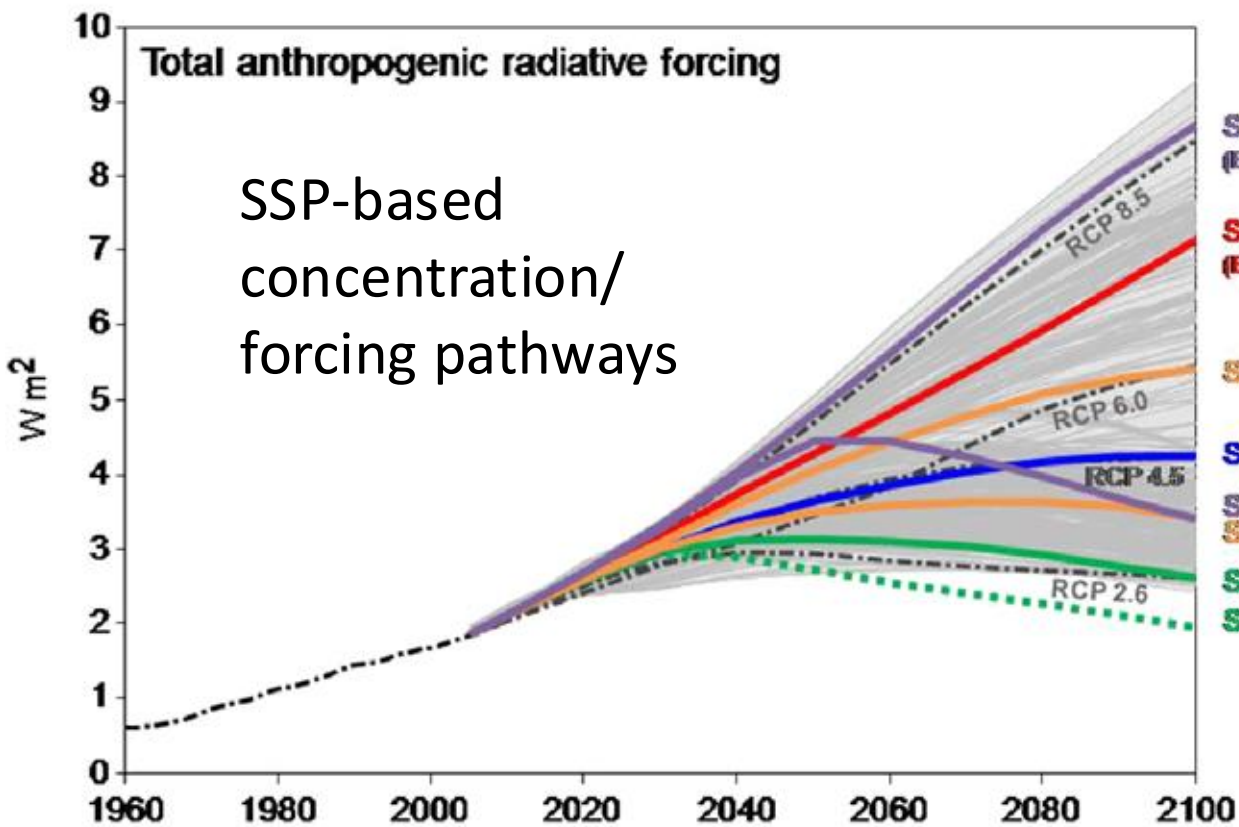
SSP Element	SSP 1			SSP 2			SSP 3			SSP 4			SSP 5		
	Country Income Groupings														
	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High	Low	Med	High
Conventional and Unconventional Fossil Fuel Conversion (synfuel and syngas in parenthesis if different)															
Technology Development	Med			Med			Low			Low	Med	Med	Med (High)		
Social Acceptance	Low			Med			High			High	Low	Low	High		
Commercial Biomass Conversion															
Technology Development	High			Med			Low			High	High	High	Med		
Social Acceptance	Low			Med			High			High	High	High	Med		
Non-bio Renewables Conversion															
Technology Development	High			Med			Low			High	High	High	Med		
Social Acceptance	High			Med			Med			High	High	High	Low		
Nuclear Power															
Technology Development	Med			Med			Low	Low	Med	High	High	High	Med		
Social Acceptance	Low			Med			High	High	High	High	Med	Med	Med		
CCS (under climate policy only)															
Technology Development	Med			Med			Med			High	High	High	High		
Social Acceptance	Low			Med			Med			High	Med	Med	Med		

SSP-based climate projections

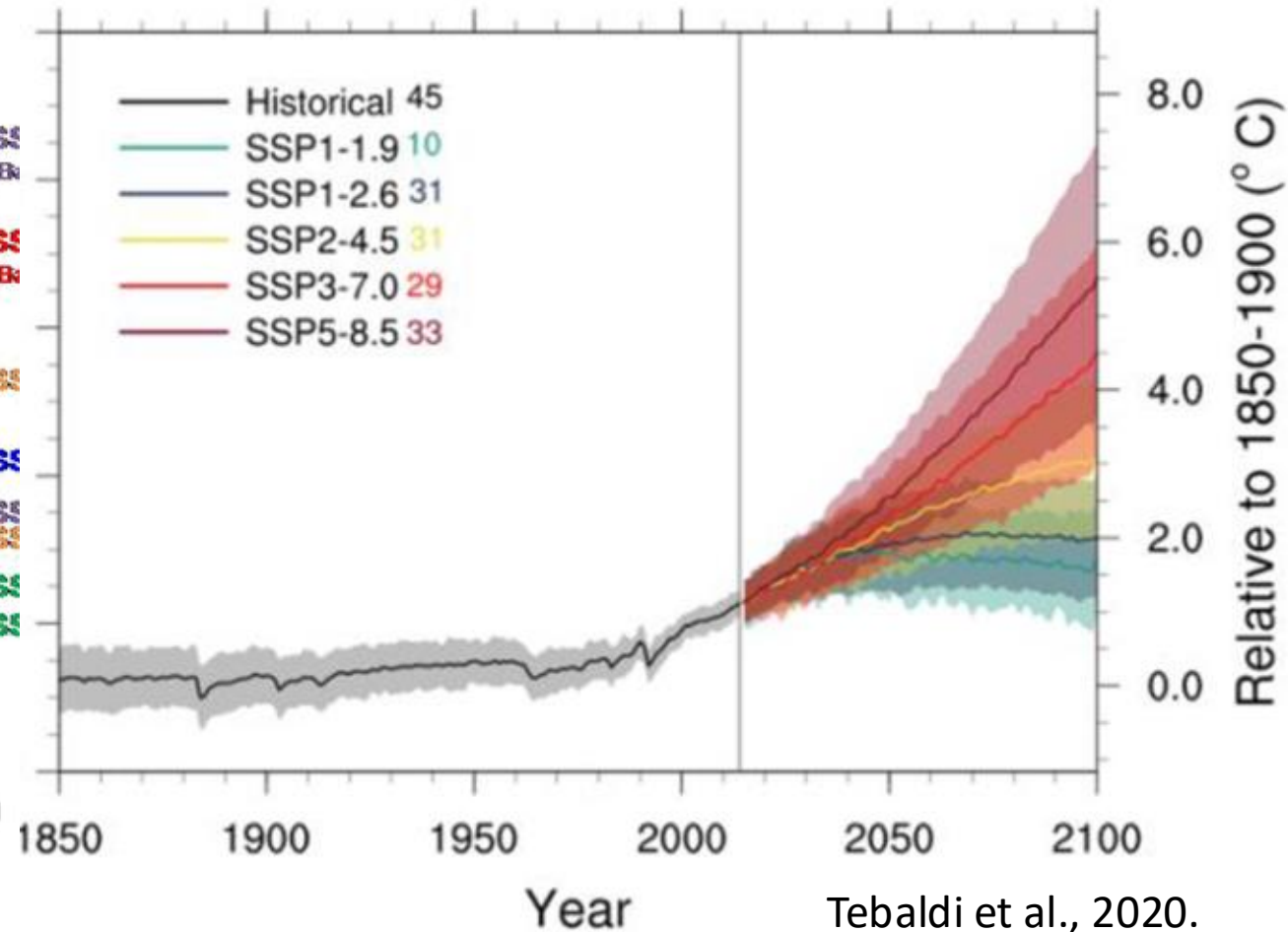
Radiative forcing
(from concentrations)



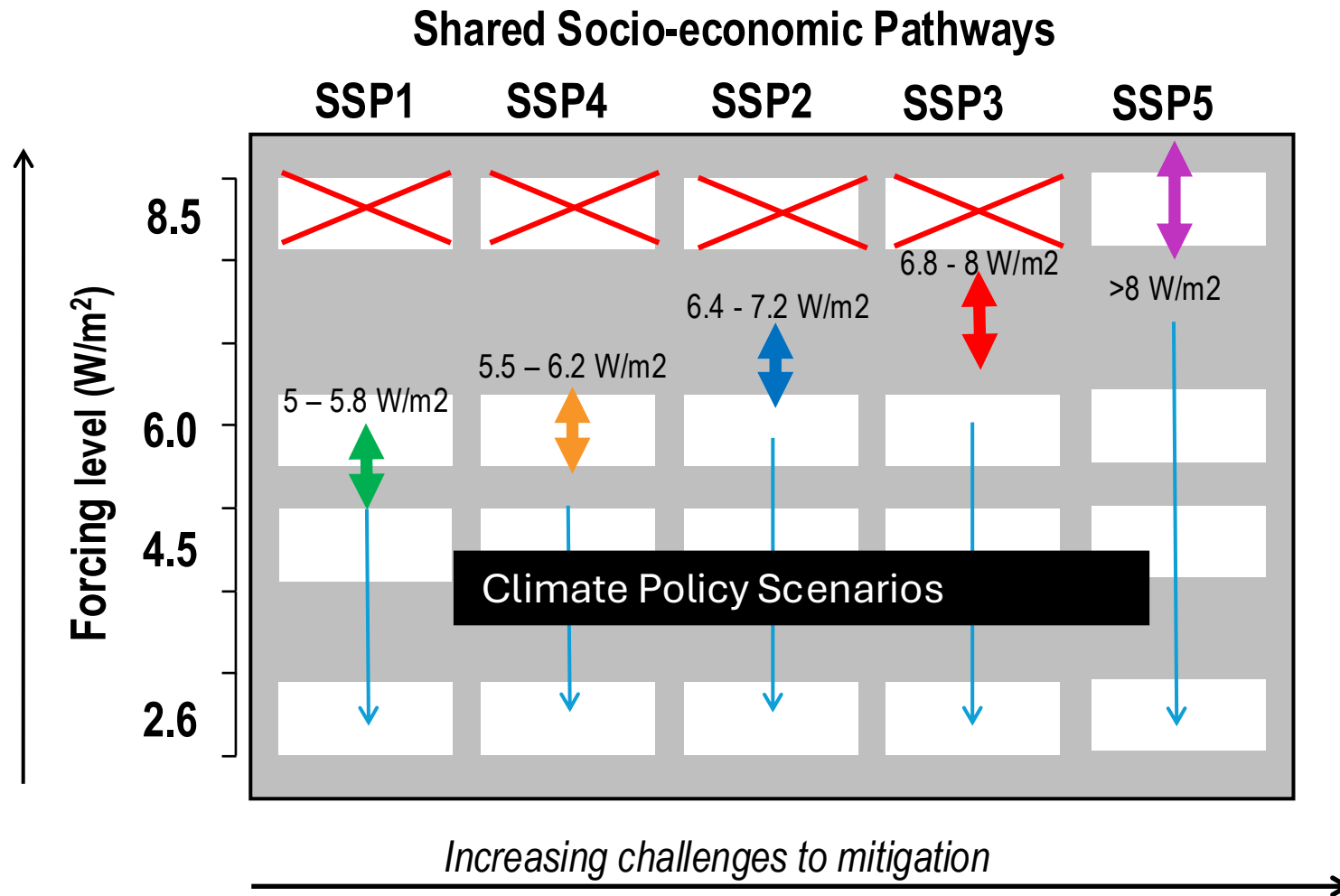
Global Mean Temp. Change
(CMIP6)



O'Neill et al., 2016; Riahi et al., 2017.

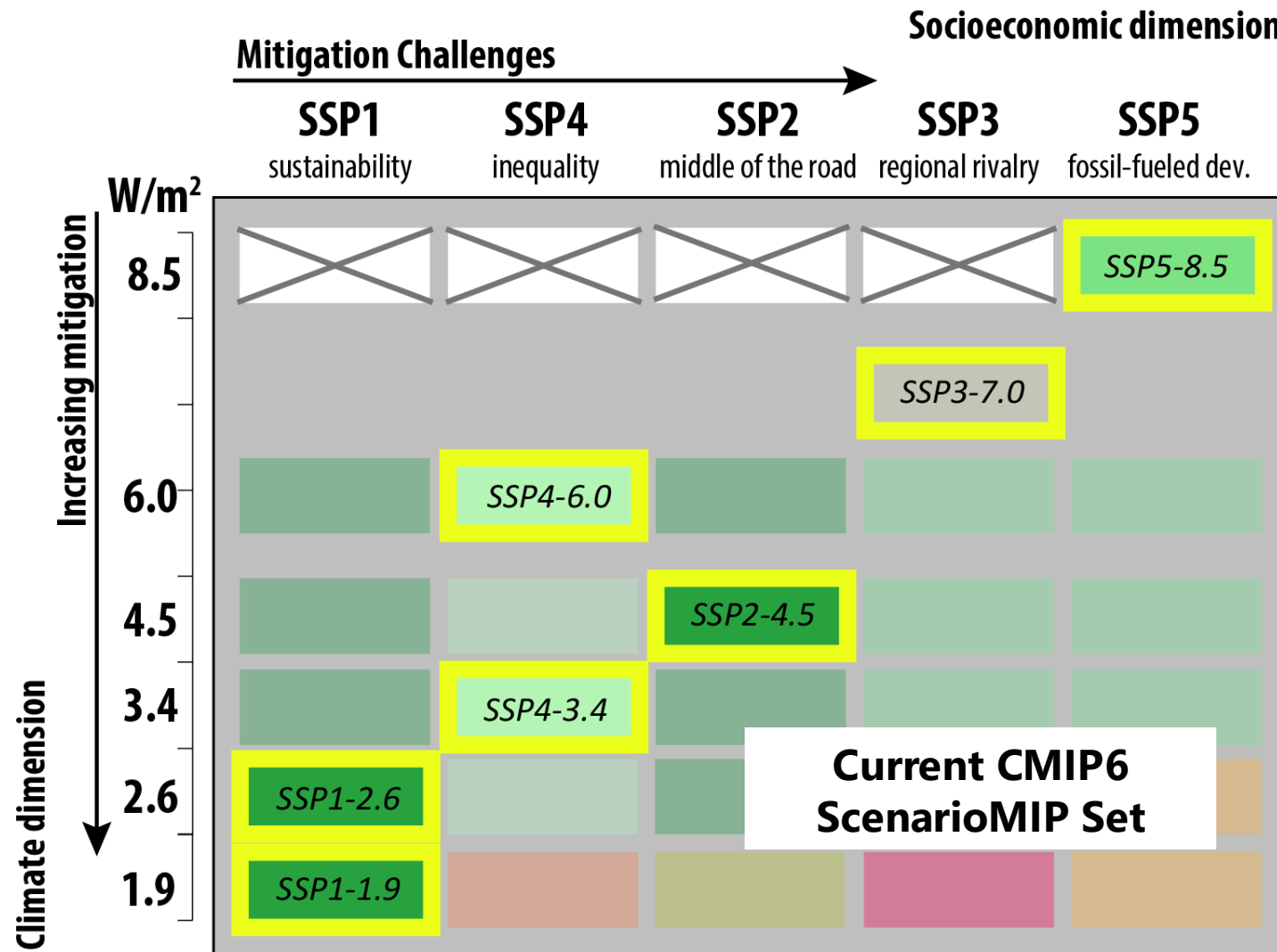


SSP/RCP combinations based on reference IAM scenarios



RCP-SSP Matrix including mitigation pathways down to 1.9W/m²

- X = number of models (or IAMs/ESMs) that successfully produced results for that SSP–RCP combination.
- Y = total number of models available for that SSP–RCP combination.



Guess the SSPx-RCPx

Conflicts such as Russia–Ukraine and Israel–Gaza drag on, making global cooperation weak. Countries like India, Pakistan, and parts of Africa rely heavily on coal and oil. Renewables grow only slowly. Warming climbs toward 4–5 °C.

Guess the SSPx-RCPx

The US and the Middle East lag behind on climate action, while the EU and Asian countries (China, Japan, South Korea) move ahead with strong renewables programs. Fossil fuels are still important in some regions, and global warming reaches about 3 °C.

Guess the SSPx-RCPx

By 2050, most people drive electric vehicles, solar and wind dominate electricity, and international cooperation is strong. Global warming is kept around 2 °C ?

Guess the SSPx-RCPx

AI and carbon capture technologies develop rapidly in the US, EU, and China, but poorer regions like Sub-Saharan Africa and South Asia are left with outdated fossil systems. Global warming lands around 3 °C.

Guess the SSPx-RCPx

China expands coal plants, Middle Eastern countries double down on oil and gas, and US shale grows. At the same time, solar and wind grow too, but lifestyles remain very energy-hungry. Warming reaches 4–5 °C.

Guess the SSPx-RCPx

India and Pakistan start cooperating on solar trade and water management, but global conflicts (Russia–Ukraine, US–China tensions) continue. Renewables grow in South Asia, but global emissions remain high.

Guess the SSPx-RCPx

Another pandemic style disruption hits in the 2030s. The EU and Asia bounce back with strong green recovery, but the US and some developing countries double down on fossil fuels to rebuild economies.

Which SSP-RCP should a planner use?

Three questions decide which scenarios to plan for



What is at stake?

- If failure would be catastrophic, as for Tarbela or the M-2 motorway, stress-test against a high-end scenario.
- If failure is cheap, a central scenario is enough.

SSP1-2.6 Paris-aligned lower bound



How long will it last?

- Before about 2040 the scenarios barely differ, so any SSP beats the historical record.
- After 2050 they diverge and the choice starts to matter.

SSP2-4.5 Close to current policies



What is it for?

- Adaptation planning should span a range of futures.
- Research that needs a clear signal above natural variability can use SSP5-8.5.

SSP3-7.0 / 5-8.5 High-end stress test